

On Nascent Angular-Momentum Distributions in Complex-Forming Bimolecular Reactions

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A detailed modelling of a complex-forming bimolecular reaction requires the knowledge of the nascent energy and angular-momentum distribution of the intermediate complex. A method is presented to calculate the distribution of the orbital angular momenta as a result of reactive collisions in a radial potential. For the case of thermal collisions, analytical solutions are obtained. Reactions of photochemically generated species are also considered. The method is applied to the reaction of CH_4 with O_2^+ and $\text{O}(^1\text{D})$, respectively.

1. Introduction

A complex-forming bimolecular reaction starts with the formation of a bound complex by a capture process. Subsequently this excited molecule may dissociate back to the reactants, may decompose or isomerize to other products, or may be stabilized by collisions. Such reactions widely occur in combustion systems [1] or atmospheric processes [2], and, for a proper modelling, an adequate description of their complicated temperature and pressure dependence is required [3]. This is most rigorously accomplished by setting up and solving a master equation [4]. In many of these reactions, angular-momentum effects can play an important role [3]. Thus, ultimately, a treatment which explicitly considers energy and angular-momentum dependence is indicated. For the reaction sequence (* formally stands for excitation)



the master equation for the concentration of C, $n(E, J)$, under steady-state conditions [5], reads as:

$$\begin{aligned} \frac{dn(E, J)}{dt} &= R_1 g(E, J) - \omega n(E, J) \\ &+ \omega \iint P(E, E'/J, J') n(E', J') dE' dJ' \\ &- [k_{-1}(E, J) + k_3(E, J)] n(E, J) = 0. \end{aligned} \quad (4)$$

Here E is the energy in the active degrees of freedom of C and J the quantum number of the total angular momentum. R_1 is the rate of reaction (1), ω the collision frequency, and $P(E, E'/J, J')$ the collisional transition probability from a state $\{E', J'\}$ to the state under consideration, $\{E, J\}$. $k_i(E, J)$ denotes the specific rate constant of the reaction (i). By solving Eq. (4), one obtains $n(E, J)$, and suitable averaging yields all quantities of interest such as branching ratios or pressure-dependent rate constants. Often one encounters experimental or practical situations, where the pressures are low, such that $\omega \ll k_i(E, J)$. Then Eq. (4) simplifies to

$$n(E, J) = \frac{g(E, J)}{k_{-1}(E, J) + k_3(E, J)}, \quad (5)$$

and an explicit solution of the master equation is not necessary.

In the present work, we concentrate on the quantity $g(E, J)$ which is the normalized nascent population of molecules C, generated by the reaction (1). From the independence of E and J follows separability:

$$g(E, J) = F(E)f(J). \quad (6)$$

For the calculation of $F(E)$, easily applicable methods are available for thermal systems [4, 6]. The situation is somewhat different, however, for $f(J)$, the normalized nascent angular-momentum distribution. Even though it is implicitly contained in thermal capture rate constants and the corresponding cross sections [7–9], there is no explicit expression available to be used in Eqs. (4) or (5). Thus we derive in the following a simple method to estimate this quantity for thermal and certain nonthermal conditions.

2. General Approach

The total angular momentum in a bimolecular capture process $\text{A} + \text{B}$ is composed from the angular momenta of the molecules A and B and from the orbital angular momentum with the quantum numbers j_A , j_B , and l , respectively. The different possible couplings are discussed in Refs. [10, 11]. Frequently one has situations, where j_A , $j_B \ll l$, and, thus, we make the general approximation $J \approx l$ and hence $f(J) = f(l)$. We believe that this is a reasonable assumption, especially at not too high temperatures and in cases, where one reactant is translationally hot. Moreover, the probability densities of j_A and j_B often can be well characterized. Then, knowing $f(l)$, one can derive the distribution function of J by a corresponding convolution under observation of the Clebsch-Gordan relation [12].

A bimolecular collision is characterized by the impact parameter b and the relative translational energy in the center-of-mass coordinate system, E_t . For a given pair of b and E_t , l follows in a semiclassical way [13] from

$$[l(l+1)]^{1/2} = \frac{(2\mu)^{1/2}}{\hbar} b E_t^{1/2}, \quad (7)$$

where μ is the reduced mass of the colliding pair, and $\hbar = h/(2\pi)$ with Planck's constant h . Eq. (7) also governs the distribution function $f(l)$ for given distributions $f_1(b)$ and $f_2(E_t)$. If N is the number of bimolecular collisions, it follows for the probability density

$$\frac{dN_{b \dots b+db, E_t \dots E_t+dE_t}}{N} = f_1(b) f_2(E_t) db dE_t. \quad (8)$$

By transforming the independent variables b and E_t into b and l , using Eq. (7), one obtains

$$\frac{dN_{l \dots l+dl}}{N} = \int_0^{b_{\max}(l)} f_3(b, l) db dl \equiv f(l) dl. \quad (9)$$

In this way, $f(l)$ can be calculated for any distribution functions $f_1(b)$ and $f_2(E_t)$. The maximum impact parameter $b_{\max}$ as an upper integration limit ensures that only reactive collisions are counted. It is connected to the reactive cross section σ_r by [13]

$$\sigma_r = \pi b_{\max}^2, \quad (10)$$

i.e. capture can only occur for $b \leq b_{\max}$. The cross section σ_r and, thus, $b_{\max}$ can alternatively be expressed as functions of l or E_t [13]. The mutual conversion is achieved by using Eq. (7).

The energetic distribution function $f_2(E_t)$ is separately discussed below for thermal and nonthermal conditions. The distribution function for the impact parameter, $f_1(b)$, however, is general and follows from the equipartition of the collisions over the whole reactive cross section [13]:

$$\frac{dN_{b \dots b+db}}{N} = f_1(b) db = \frac{2b}{b_{\max}^2(E_t)} db, \quad (11)$$

where $b_{\max}$ contains the complete information on the intermolecular potential.

3. Maximum Impact Parameters from Intermolecular Potentials

Capture processes are often governed by the long-range part of the intermolecular potential. In this region, the latter is nearly isotropic and can be approximated by its dominant electrostatic term in the form

$$V(r) = -\frac{C_s}{r^s} \quad (12)$$

with $s > 2$ and r being the distance between the colliders. For the different contributions to C_s , see e.g. [14]. In an isotropic potential, l is conserved, and $b_{\max}$ can be derived from the maximum of the effective potential $V_{\text{eff}}(r) = V(r) + E_t b^2/r^2$ [13]. One obtains

$$b_{\max}(l) = \left\{ \left(\frac{2}{s} \right)^{s/2} \left(\frac{s-2}{2} \right)^{(s-2)/2} \frac{\hbar^2}{2\mu C_s} [l(l+1)] \right\}^{1/(2-s)}. \quad (13)$$

If the use of Eq. (12) is not appropriate, and/or an experimental capture rate constant, k_{capt} , is available, a rough estimate of $b_{\max}$ also follows from

$$b_{\max} = \left(\frac{k_{\text{capt}}}{\langle v \rangle \pi} \right)^{1/2} \quad (14)$$

with the average relative velocity $\langle v \rangle = (8k_B T/(\pi\mu))^{1/2}$, where k_B is Boltzmann's constant and T the temperature. This latter approach corresponds physically to the very simple picture of reactive hard spheres with no activation energy [15].

In a somewhat different but more rigorous way [7–9], the capture cross section, and via Eq. (10) also $b_{\max}$, is related to the number of open reaction channels, $W(E, J)$, by

$$\sigma_r(E, E_t, J) = \frac{\pi \hbar^2}{2\mu E_t} (2J+1) W(E, J). \quad (15)$$

Here the intermolecular potential is implicitly contained in $W(E, J)$, and, in principle, potentials of any complexity can be handled. A number of counting procedures or approximations for the determination of $W(E, J)$ exist (see below), and the integration over E_t finally leads to $g(E, J)$. Nevertheless we will not trace this path further in the present work, because at this stage only the main effects shall be demonstrated, and easily applicable formulas, without the necessity of determining $W(E, J)$, shall be provided.

4. Angular-Momentum Distributions

4.1 Thermal Collisions

The distribution $f_2(E_t)$ for the relative translational energy in thermal collisions reads as [13, 15]:

$$\frac{dN_{E_t \dots E_t+dE_t}}{N} = f_2(E_t) dE_t = \frac{1}{(k_B T)^2} E_t \exp\left(-\frac{E_t}{k_B T}\right) dE_t. \quad (16)$$

By employing this function together with Eqs. (11) and (7), Eq. (8) can be transformed into Eq. (9). Then, using $b_{\max}$ from Eq. (13), the integration with respect to l can be carried out analytically. With the abbreviation $y = [Dl(l+1)]^{1/2}$, one obtains a very compact uniform function

$$\frac{dN}{N} = 2y \Gamma(1+2/s) \mathcal{Q}[1+2/s, y^{2s/(s-2)}] dy, \quad (17)$$

from which $f(l)$ can be easily extracted. Here $D = \hbar^2/[(k_B T)^{1-2/s} 2\mu A^2]$, where A contains the potential parameters and is given in Table 1. Γ denotes the common and $\mathcal{Q}$ the incomplete gamma function, for which convenient routines exist [16]. The limiting cases for $s \rightarrow 2$ and $s \rightarrow \infty$ can also be derived and are included in Table 1. The latter case corresponds to the hard-sphere model and the use of $b_{\max}$

from Eq. (14). Average values, $\langle y \rangle$, to obtain $\langle l \rangle$ are also given in Table 1, and plots of Eq. (17) for different values of the parameter s are shown in Fig. 1.

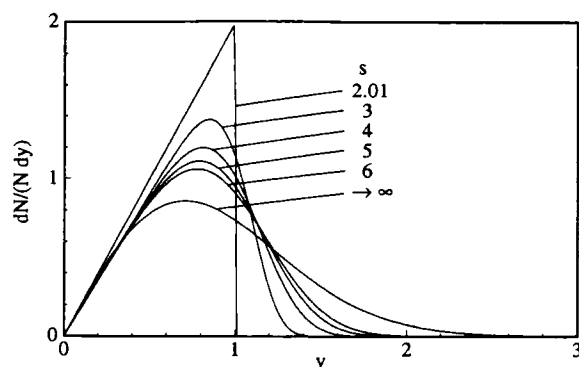


Fig. 1
Probability densities according to Eq. (17). The curve for $s = 2.01$ is included to demonstrate the correct limiting behavior for $s \rightarrow 2$

4.2 Nonthermal Collisions

In experimental investigations of complex-forming bimolecular reactions, often one reactant is generated by photolysis of a suitable precursor. The reaction scheme (Eqs. (1) – (3)) then has to be extended by a step $\text{XA} + h\nu \rightarrow \text{X} + \text{A}$, where A, often, is an atom and carries an excess of translational energy, E_{exc} . Based on earlier works of Bernstein [17] and Chantray [18], van der Zande et al. [19] emphasized the superthermal width of the energy distribution in the subsequent collisions $\text{A} + \text{B}$. From this distribution (Eq. (3) of Ref. [19]) follows

$$f_2(E_t) = F_n E_t^{1/2} \{ \exp [-(u_1 - u_2 E_t^{1/2})^2] - \exp [-(u_1 + u_2 E_t^{1/2})^2] \} \quad (18)$$

with the normalization factor $F_n = u_2^3 / [2 u_1 \exp(-u_1^2) + \pi^{1/2} (1 + 2 u_1^2) \text{erf}(u_1)]$. Furthermore, $u_1 = [m_X m_B E_{\text{exc}} / (m_A m_{\text{ABX}} k_B T)]^{1/2}$ and $u_2 = [m_{\text{AX}} m_{\text{AB}} / (m_A m_{\text{ABX}} k_B T)]^{1/2}$, where m_i is the mass of the molecule/atom i , and $\text{erf}(z)$ the error function [16]. The energy E_{exc} , to a first approximation, can be identified with the difference between the photon energy and the bond-dissociation energy of X-A [19]. With the distribution from Eq. (18), the integration in Eq. (9) again can be carried out analytically for certain values of s . However, the expressions become very complex except for the hard-sphere potential with $s \rightarrow \infty$. In this case, using $\sigma_r = k_{\text{capt}} / \langle v \rangle$, one obtains

$$\frac{dN_{l \dots l+dl}}{N dl} = \frac{(2l+1) \pi^{1/2} G^2 F_n}{u_2^3} [\text{erf}(u_1 - G l^*) + \text{erf}(u_1 + G l^*)], \quad (19)$$

where $G = [\pi \hbar^2 m_{\text{AX}} m_{\text{AB}}^2 / (2 \sigma_r k_B T m_A^2 m_B m_{\text{ABX}})]^{1/2}$, and the abbreviation $l^* = [l(l+1)]^{1/2}$ is employed. One should note that Eq. (19) is independent of u_2 .

Using Eq. (19), one can easily account for the main effects of the broad distribution (Eq. (18)) even though the hard-sphere capture cross section is surely a very simple model. If more detailed informations on the potential-energy surface are available, they may be included into $b_{\text{max}}(l)$, and Eq. (9), then, would have to be integrated numerically.

5. Applications and Discussion

For the sake of demonstration, we apply our method at first to the thermal capture process $\text{O}_2^+ + \text{CH}_4$ which can be described by an isotropic potential-energy surface. The overall reaction, however, is very complicated [20], and rotational effects were shown to play an important role [21]. Thus, for a detailed modelling, information on the angular-momentum distribution is essential. With the isotropic polarizability of CH_4 , $\alpha = 2.593 \cdot 10^{-24} \text{ cm}^3$ [22], it follows for the charge/induced-dipole potential [14] $C_s = \alpha e^2 / 2 = 1.51 \cdot 10^5 \text{ cm}^{-1} (10^{-10} \text{ m})^4$, where e is the electron charge. Then, for $s = 4$, one obtains, from Table 1, $A = 27.9 \cdot 10^{-10} \text{ m} (\text{cm}^{-1})^{1/4}$ and, thus, $D = 1.41 \cdot 10^{-4}$. The resulting distribution for $T = 300 \text{ K}$ is shown in Fig. 2. Also included is a distribution which was derived using a hard-

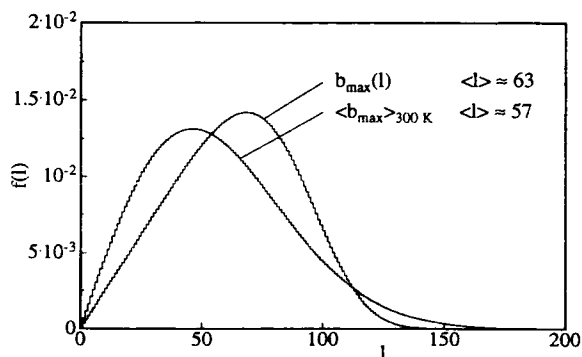


Fig. 2
Probability density $f(l)$ for the thermal capture process $\text{O}_2^+ + \text{CH}_4$ and two different models for the capture cross section (see text). $\langle l \rangle$ denotes the average angular momentum quantum number

Table 1
Distribution functions and parameters; for explanations, see text

s	A	$dN/(N dy)$	$\langle y \rangle$
∞	$[k_{\text{capt}} / (\langle v \rangle \pi)]^{1/2}$	$2y \exp(-y^2)$	$\pi^{1/2} / 2$
2	$C_2^{1/2}$	$2y$ with $0 \leq y < 1$	$2/3$
≥ 3	$(s/2)^{1/2} [2/(s-2)]^{(s-2)/(2s)} C_s^{1/s}$	$2y \Gamma(1+2/s) Q[1+2/s, y^{2s/(s-2)}]$	$2\Gamma(5/2-1/s)/3$

sphere potential, i.e. $s \rightarrow \infty$, with a constant $\langle b_{\max} \rangle$. The latter was obtained by averaging the actual $b_{\max}(E_t)$ over $f_2(E_t)$ from Eq. (16). As one can see, this approximation is not too bad, especially with regard to the average values, $\langle l \rangle$.

For this type of reaction, exact channel potentials and the corresponding maxima are given in Ref. [23]. They follow as a limiting case for ion-molecule reactions with isotropic polarizabilities and vanishing higher multipole moments. Together with a suitable counting procedure, they may be used in Eq. (15). In Ref. [23], also thermal as well as state-resolved capture rate constants are derived, which again implicitly contain $f(l)$. However, no explicit expressions are available.

As a second example, we consider the reaction $O(^1D) + CH_4 \rightarrow CH_3OH^*$, which has drawn a lot of attention over the last years under several points of view (see e.g. Ref. [24] and the literature cited therein). A first experimental determination of the lifetime of the intermediary methanol was done by Lin and DeMore [25]. In their work, $O(^1D)$ was generated from the photolysis of N_2O by a mercury lamp at 184.9 nm with CH_4 as bath gas. These experiments shall provide the frame for our examples: we calculate the nascent methanol populations for the two limiting cases, i.e. for a translationally hot $O(^1D)$ from the photolysis and for a situation, where translation is thermalized before reaction occurs. Because only the gross features shall be demonstrated, a constant, energy-independent, reactive cross section is sufficient.

From the experimental rate constant $k_{\text{capt}} = 1.4 \cdot 10^{-10} \text{ cm}^3 \text{ s}^{-1}$ [26] follows, with $T = 300 \text{ K}$, a reactive cross section $\sigma_r = 15.7 \cdot (10^{-10} \text{ m})^2$. Employing standard thermochemistry [22, 26] for the photolysis step $N_2O \rightarrow N_2 + O(^1D)$ and the above wavelength, one obtains $E_{\text{exc}} = 24300 \text{ cm}^{-1}$. Finally, these values together with the masses of the corresponding fragments yield $u_1 = 7.38$ and $G = 0.0543$. The resulting distribution according to Eq. (19) is shown in Fig. 3a.

For the thermal case, inspection of Table 1, for $s \rightarrow \infty$ and $T = 300 \text{ K}$, yields $A = 2.24 \cdot 10^{-10} \text{ m}$ and, thus, $D = 2.02 \cdot 10^{-3}$. One should note that this distribution, which is also contained in Fig. 3a, completely coincides with a thermal rotational distribution of the corresponding quasi-diatomic molecule with an interfragment distance $b_{\max} = A$. This, additionally, demonstrates the correct limiting behaviour of Eq. (17) for $s \rightarrow \infty$.

In Fig. 3b, also the energy distributions for the two cases are shown. The thermal curve was obtained from the usual statistical approach (see e.g. Eq. (9–36) of Ref. [6]), where the molecular data for methanol stem from Ref. [27]. The other distribution, resulting from the translationally excited $O(^1D)$, was estimated by convoluting Eq. (18) with a thermal population of methanol calculated from the densities rather than the sums of states [6].

The two methanol populations, of course, show different lifetimes. Using the statistical adiabatic channel model (SACM) [28], one calculates values of $\approx 10 \text{ ps}$ and $\approx 1.6 \text{ ps}$ for CH_3OH^* from thermal and translationally hot $O(^1D)$,

respectively. This indicates difficulties in comparing lifetimes from different investigations.

Lin and DeMore [25] carried out their experiments in CH_4 as bath gas. Because the rate constant for $CH_4 + O(^1D)$ is near the gas kinetic collision limit, the methanol population may correspond to the hot distribution considered above. The lifetime they determined in a relative way is $\approx 0.8 \text{ ps}$. As reference, a strong-collision model for the stabilization was employed. However, if one takes into account a collisional efficiency < 1 , the lifetime will increase and may well fall into the range of the statistical prediction. Thus, it seems questionable to take this value of 0.8 ps as an indication per se for the nonstatistical character of the methanol decomposition from $O(^1D) + CH_4$. Also, in a recent experiment by van Zee and Stephenson [29], a somewhat larger lifetime of $\approx 3 \text{ ps}$ was determined by photolysis of a $CH_4 \cdot O_3$, van der Waals complex. Thus, by not taking into account the high translational energy of $O(^1D)$, one would get a much larger discrepancy between experiment and statistical approach, notwithstanding the question, if statistical theory is really adequate [24, 30]. The problem of translational excitation, of course, disappears in systems with an inert bath gas. Because the translation is equilibrated by only a few collisions, the thermal distribution (Eq. (17)) surely applies.

Finally, we will discuss some additional aspects regarding the implementation of more realistic potential-energy surfaces. Our approach so far, i.e. to consider only the radial

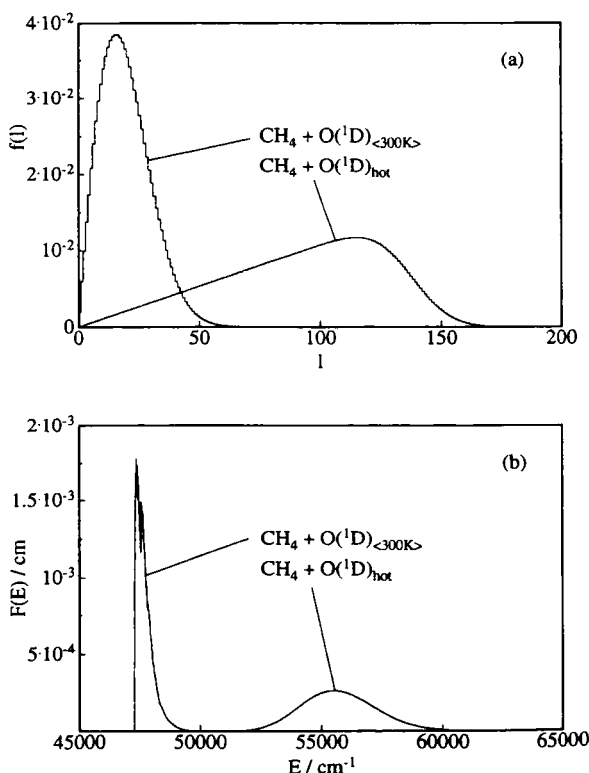


Fig. 3 Nascent angular momentum (a) and internal energy (b) distribution of CH_3OH formed by $CH_4 + O(^1D)$

part of the potential, is known as phase-space theory (PST). By taking into account also the anisotropy, one introduces "rigidity" in a sense that the maxima of the channel potentials are shifted towards lower values of r [31]. Hence the transition states have more product-like properties (seen from the bimolecular direction), and in this way, also $b_{\max}$ is decreased. One way to include these effects is provided by the statistical adiabatic channel model. As was already mentioned in Sect. 3, the bimolecular capture cross section can be related to the number of open reaction channels, $W(E, J)$ [7–9]. This offers a very convenient possibility to account for the anisotropies by means of rigidity factors introduced by Troe [31–33]. In order to obtain the correspondingly altered cross sections, one simply has to multiply $W_{\text{PST}}(E, J)$ for the phase-space limit by a specific rigidity factor. For the determination of $W_{\text{PST}}(E, J)$, numerical as well as rapid approximate methods exist [11, 34–39]. If the coupling between the orbital and reactant angular momenta can be treated correctly, also the approximation $J \approx l$ can be dropped. Nevertheless, one still has to do the averaging over the different distributions $f_2(E_t)$, and an explicit elaboration to yield $g(E, J)$ or $f(J)$ has not yet been done. With this in mind, and considering the lacking knowledge on most potential-energy surfaces of chemical interest, we believe that our method, because of its compactness, may be useful for many practical applications. A more rigorous treatment, of course, has to include the complete potential and may follow the way just outlined.

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